Table 1: HyperKKL rank ablation at the matched 140k budget (dev systems; the trunk width is re-solved per rank: $w = 149, 44, 11$ for $r = 1, 4, 16$, Table **??**). Absolute NRMSE and paired change vs $r = 4$ (seeds 0-2, ± 95% CI half-width; bold/underline: significant). Pre-registered rule (C1-rank1/16): within ±10% of $r = 4$ or better (TOST non-inferiority, 90% CI) on ≥4/6 systems, sequence OOD.

tier	system	$r = 1$	$r = 4$	$r = 16$	$r = 1$ vs 4	$r = 16$ vs 4
shape OOD	battery	0.0715	0.0551	0.0555	+33% ±111	+3% ±84
	bioreactor	0.0257	0.0239	0.025	+8% ±73	+6% ±50
	duffing	0.0522	0.0518	0.0477	+1% ±26	-8% ±9
	lorenz	0.0406	0.0257	0.0232	+58% ±18	-10% ±21
	quad_tank	0.0556	0.0502	0.0558	+11% ±13	+11% ±2
	vdp	0.0198	0.0158	0.0162	+25% ±20	+3% ±9
sequence OOD	battery	0.0904	0.0719	0.0768	+29% ±112	+12% ±135
	bioreactor	0.0631	0.0617	0.0694	+1% ±42	+10% ±108
	duffing	0.0581	0.057	0.0529	+2% ±26	**-7%** ±6
	lorenz	0.0421	0.0266	0.0255	+58% ±32	-4% ±16
	quad_tank	0.0721	0.07	0.0742	+3% ±7	+6% ±10
	vdp	0.0235	0.0191	0.0195	+23% ±4	+2% ±7

C1-rank1: FAIL – lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 1/6 (need 4/6)
C1-rank16: FAIL – lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 3/6 (need 4/6)